Human Ancestors- Transition to Modern Humans



Homo heidelbergensis
Homo neanderthalensis
Other extinct recent hominins
Homo sapiens

Homo heidelbergensis / antecessor

- Possible (likely?) common ancestor of both *Homo sapiens* and *Homo neanderthalensis*
- Fossils
  - found in Africa (dated to 1 mya)
  - Found in Europe (dated to 500 000 years ago)
- Features between *H. erectus* and *H. sapiens*
- Not enough fossil evidence to fully understand relationships between different *Homo* species.
  - Family tree has more branches than at first thought – just all extinct.

One hypothesis:

- *H. heidelbergensis* populations in Europe in cold, extreme conditions due to ice age. Certain body type favoured - evolved into *H. neanderthalensis*.
- *H. heidelbergensis* populations in Africa had different selection pressures – ice age caused low rainfall, dry environment – poor access to water and food. Problem-solving and high cognitive capacity were a selective advantage – *Homo sapiens* evolved.

Learning Aim: Describe the likely evolutionary pathway that led to the evolution of H.neanderthalensis and H.sapiens

Homo heidelbergensis



Relationship to humans:

- Direct ancestor of humans, also of Neanderthals

Time Period:

- 200 000—700 000 years ago—Pleistocene

Distribution:

- East Africa, migrating out to Europe and part of Asia



Climate:

- Earth moving into an ice age. Low sea levels allowed movement across land bridges

Fossil Record and what it tells us:



- Variety of fossils found in Africa and Europe show:
  - Tall and long legged.
  - Stongly built
  - Prominent brow ridges
  - Larger brain than previous species—1250 cc. Frontal and parietal lobe enlargement indicates complexity.
  - Tooth size decreasing—closer to modern humans
  - Face more flattened than previous species.
  - No pointed chin
  - Parabolic dental arcade
- Stone tools—used Acheulian tools similar to those used by *H. erectus*
- Evidence of hearths showing fire use, and butchery of animals for food.



Cultural Evolution:

Tool Use

- Made hand-axes "Acheulian". Finely flaked, bifacial tool. Used for chopping , cutting.
- Wooden spears for hunting



Fire Use

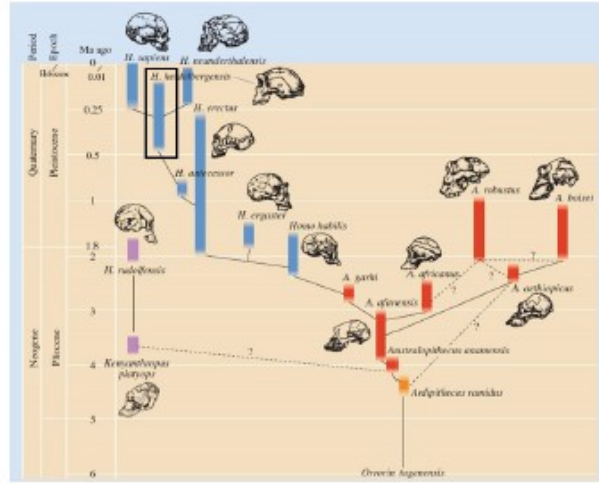
- Evidence of fire use in hunting, and cooking. Likely able to make and control fire

Language, Art, Abstract Thought

- Likely to have had language, growing in complexity
- No evidence of art or abstract thought, but evidence of some ability to plan.

Built Structures

- Made shelters

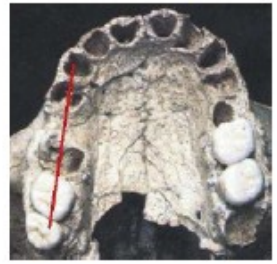


Lifestyle:

- Modified environment to suit them
- Bipedal gait
- Family groups
- Made shelters and used tools
- Cared for elderly and disabled
- Planned hunting including spearing large herd animals.

Diet and Dentition:

- Large amounts of meat, supplemented with plant foods.— Allowed for brain development
- Teeth show movement towards parabolic dental arcade:
  - Teeth smaller than previous species.

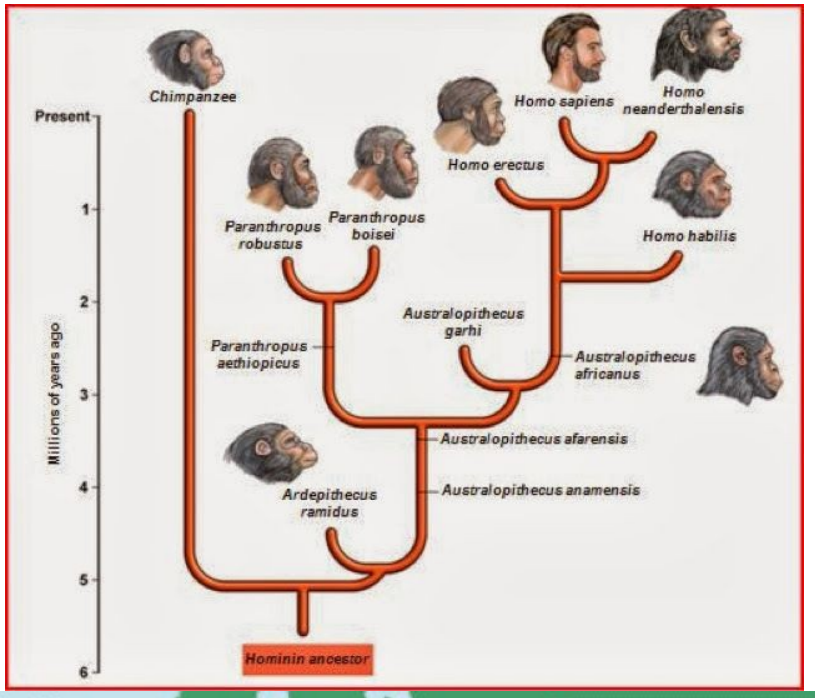


Homo neanderthalensis – The Neanderthals - Overview

- First fossils found in 1856 in Neander Valley, Germany
- Many more fossils found since.
- Side branch along pathway to modern humans – we share a common ancestor
- DNA shows a distinct biological species:

  Homo neanderthalensis

- Existed in Europe during the last ice age and were adapted for those conditions
  - 400 000 years BP to 40 000 year BP
- Neanderthals and *Homo sapiens* coexisted in Europe towards the end of the ice age, and inter-bred.





Learning Aim: Provide an overview of H.neanderthalensis including fossil record, time of existence, range and habitat.

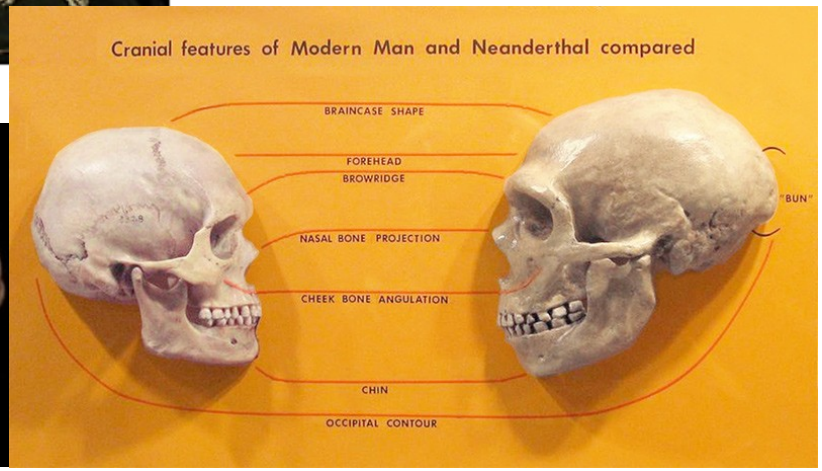
Homo neanderthalensis – Anatomical Features

- Features include:
  - Big faces
  - Low but large skulls – back of skull drawn out into "bun"
  - Cranial capacity larger than humans, but frontal cortex smaller
  - Heavy brow ridges
  - Lack of defined chin
  - Cheeks swept back
  - Somewhat prognathic jaw
  - Likely to have very prominent, wide nose – adaptation for cold, dry air and to regulate body temperature without sweating
  - Solid frame – shorter and stockier than *H.sapiens*
  - Other skeletal features very similar to *Homo sapiens*
  - Probably walked, ran and used hands in way very similar to modern humans









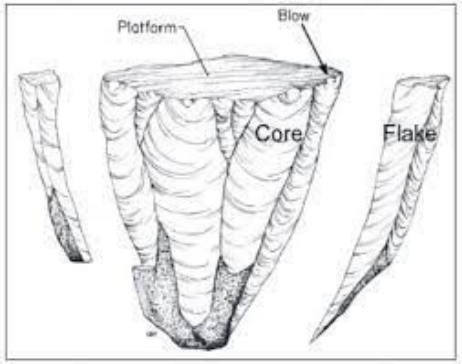
Learning Aim: Describe anatomical features of H. neanderthalensis

Homo neanderthalensis – Lifestyle and Culture

- Specialised hunters
  - Meat an important part of diet
  - Plant food sources scarce in ice-bound Europe
  - Needed energy from animal fats to survive the cold
- Made sophisticated stone tools
  - "Mousterian Industry"
  - Flaked a range of tool types from stone cores – many tools from a single core
  - Hafted (stuck) stone tools to wood to make axes, spears.
  - Made stone scrapers to scrape hides, then pierced them with sharp tools and bound them together with strips of hide to make clothing.
  - Previous hominins made simple, generalised tools and did not use the core and flaking technique to make many different tools from a single stone
- Made clothing
- Likely to have lived in caves
- Could make and use fire for hunting, cooking and warmth
- Buried dead, cared for sick and injured
- No evidence of abstract thought or symbolic representation eg cave art.







Learning Aim: Describe lifestyle and culture of H.neanderthalensis

Homo neanderthalensis – Lifestyle and Culture





Learning Aim: Describe lifestyle and culture of H.neanderthalensis

Homo neanderthalensis



Relationship to humans:

- Common ancestor with humans.
- Evolved from geographically isolated population of *H. heidelbergensis*
- Coexisted with humans, some interbreeding

Time Period:

- 40 000—400 000 years ago

Distribution:

- Ice-age Europe and Central Asia.



Climate:

- Ice –age Europe.
- Cold, glaciated

Fossil Record and what it tells us:

- Variety of fossils found in Europe show:
  - Shorter, more robust and muscular than modern humans
  - Stongly built
  - Prominent brow ridges
  - Brain similar size to modern humans.
  - Long, low skull, rounded brain case
  - Occipital bun—strong neck muscle attachment
  - Some jaw projection
  - No pointed chin
  - Large nose
  - Teeth larger than modern humans



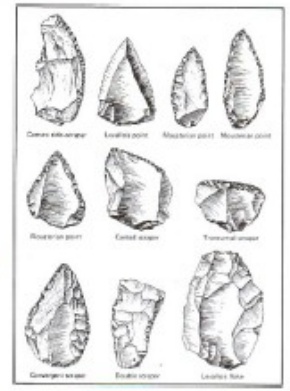


- Stone tools—used complex stone tools
- Evidence of hearths showing fire use, and butchery of animals for food, and wearing of skins as clothing.

Cultural Evolution:

Tool Use

- "Mousterian" industry—used stone core and flaked off and refined a range of tools



Fire Use

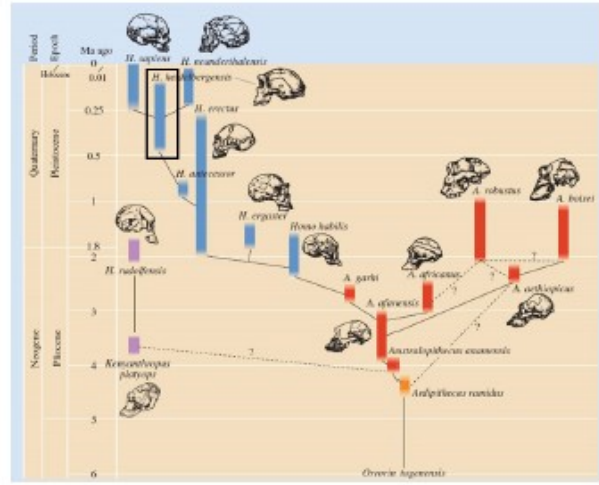
- Evidence of fire use in hunting, and cooking. Able to make and control fire

Language, Art, Abstract Thought

- Likely to have had language, growing in complexity
- Some evidence of burial of dead
- Little evidence of art or adornment

Built Structures

- Used caves but also made shelters



Lifestyle:

- Modified environment to suit them
- Bipedal gait
- Family groups
- Made shelters and used tools
- Cared for elderly and disabled
- Planned hunting—ambush of large animals

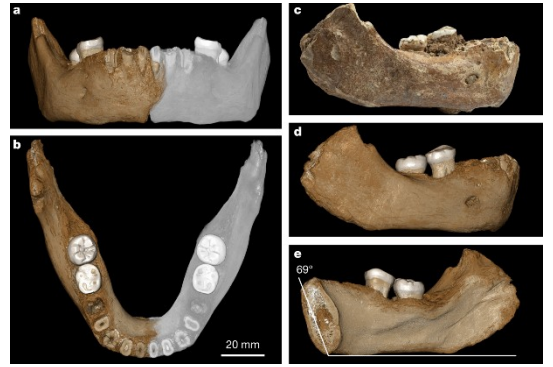
Diet and Dentition:

- Large amounts of meat, supplemented with plant foods.
- Teeth larger than modern humans
- Parabolic dental arcade



Other recent extinct hominins - The Denisovans

- Bone fragment discovered in 2008
  - Denisova cave, Siberia
  - Approx 40 000 years old
  - mtDNA not neanderthal nor modern human
  - Analysis of DNA base similarity indicates common ancestor with humans and neanderthals lived 1 million years ago.
  - Good example of how mtDNA can be used to analyse relationships between hominins.
  - Nuclear DNA shows
    - more closely related to neanderthals than modern humans
    - Have interbred with modern humans, in particular ancestors of modern Melanesians and Australian Aboriginal people
  - Not much other evidence about culture and behaviour
  - More evidence will reveal more detail, and more evidence is available all the time.







Learning Aim: Briefly describe the location, time range of existence, and features of Denisovans

Other recent extinct hominins - The Red Deer Cave People

- Skull discovered in 1979 in China
  - Thick bones
  - Prominent brow ridges
  - Short, flat face
  - Lack of chin
  - Anatomically unique – a previously undiscovered hominin
  - Evidence that they cooked lots of red deer – leading to the name
  - Unclear how they fit in with hominin evolution
  - May have evolved in East Asia in parallel with *H sapiens*.
  - May be related to Denisovans?
  - Survived until relatively recently – 11 500 years ago
  - Have not yet been able to extract enough DNA to analyse to clarify relationships





Learning Aim: Briefly describe the location, time range of existence, and features of the Red Deer Cave People

Homo sapiens – Modern Humans – overview and fossil record

- Existed 300 000 years ago to present day (and future)
- 1868: fossils found at Cro-Magnon in France
  - Five people
  - Animal bones
  - Sea shell necklaces
  - Stone tools
  - Part of widespread population in Europe from 40 000 years ago to 12 000 years ago – early modern humans



- Members of *Homo sapiens* "Cro-Magnon Man"
  - Less prominent brow ridges, flatter face, smaller jaw and teeth
  - Shorter, rounder skulls
  - Chin
  - Large brain
  - Well separated eye sockets
- Many more fossils have been found since this initial fossil discovery
- Huge amount of cultural development since first evolved



Learning Aim: Provide an overview of *H.sapiens* including the fossil record and anatomical features

Homo sapiens – Modern Humans

